

Devkumar Banerjee

(979)-326-8107 | dkbanerjee314@gmail.com | [linkedin.com/in/dvkb](https://www.linkedin.com/in/dvkb) | github.com/aptitudepi | dvxb.io

EDUCATION

Texas A&M University

College Station, TX

Bachelor of Science in Computer Science

Aug 2025 – May 2029

- **Scholarships:** Brown Foundation Scholar, National Merit Scholar, President's Endowed Scholar.
- **Honors:** Craig and Galen Brown Engineering Honors; University Honors.
- **Organizations:** Aggie Computing Club (Project Manager), Engineering Honors Executive Committee (Operations Subcommittee Vice Chair), Engineering Honors – Enhancing the Development of Global Engineers (EH EDGE), TAMU Cybersecurity Club, TAMU Emergency Care Team.
- **SAT:** 1550/1600 (Math: 800/800, EBRW: 750/800).
- **PSAT:** 1520/1520 (Full Score).

RESEARCH EXPERIENCE

Cancer Radiology Research Intern

Jun 2026 – Present

The University of Texas MD Anderson Cancer Center

Houston, TX

- Selected as 1 of 6 interns from 3,000+ applicants (Top 0.2% selectivity) for a competitive, funded residency-track research internship at the UT MD Anderson Cancer Center.
- Screened electronic health records (Epic) to build a retrospective cohort of prostate cancer patients, filtering by Decipher genomic risk scores and clinical stage boundaries.
- Co-evaluated ¹⁸F-DCFPyL (Pylarify) PSMA PET/CT molecular imaging markers alongside PI Dr. Nghi Nguyen to track precise anatomical metastatic disease progression.

Undergraduate Researcher

Jan 2026 – Present

Texas A&M University – DIVE Lab

College Station, TX

- Conducting undergraduate graph neural network research under PI Dr. Shuiwang Ji and co-PI Dr. Jean-Briac le Graverend
- Built physics-informed spatiotemporal graph neural networks in PyTorch Geometric to predict martensite/austenite phase-transition trajectories in polycrystalline shape-memory alloys, serving as a surrogate for computationally expensive crystal-plasticity finite-element simulations.
- Assisted in architecting a hybrid GNN model combining a graph backbone (GIN, then DIG SchNet with Gaussian radial-distance embeddings and shifted-softplus filters) with FiLM-conditioned physics encoders and a GRU temporal module over 50-grain, 22-feature Voronoi face-sharing graphs.
- Designed an annealed Gaussian input-feature noise regularization module that reduced one-step prediction MSE by 2.2× and validation loss by 27% over the prior model baseline.
- Conducted a 5-point σ hyperparameter sweep and a 6× data-scaling study across 1–6 loading conditions and 4 grain orientations to characterize model generalization.
- Diagnosed a "one-way ratchet" failure mode in 110-step autoregressive rollouts by parsing per-step worst-feature error logs, isolating the model's inability to shrink high martensite-fraction predictions and motivating a multi-step rollout/pushforward loss experiment.

Undergraduate Researcher

May 2025 – Present

Texas A&M University – AGGIES Lab

College Station, TX

- Conducting undergraduate research under PI Dr. Shreyas Kumar on benchmarks, datasets, and evaluation pitfalls for water treatment plant security, curating and analyzing 36 SWaT/WADI-based papers to develop taxonomies, identify methodological issues, and propose standardized evaluation guidelines.
- Co-authoring a Systematization of Knowledge manuscript currently in preparation for publication.
- Developing and evaluating machine learning pipelines on multi-modal clinical data (structured risk scores + acoustic features) as part of an international collaboration with PI Dr. Shreyas Kumar and LogicBoots Pvt. Ltd., targeting patient-specific outcome prediction within a remote-sensing IoT stethoscope device.

Machine Learning & Cancer Nanomedicine Research Intern

Mar 2023 – Aug 2025

Houston Methodist Research Institute

Houston, TX

- Selected as 1 of 15 students from 2000+ applicants (Top 0.75% selectivity) for an in-person, paid research internship at the Houston Methodist Research Institute in the Texas Medical Center.

- Collaborate with PI Prashant Dogra and Ph.D. student on nanoparticle (NP) tumor biodistribution study using statistical analysis and machine learning methods.
- Analyzed 280+ *in vivo* nanoparticle administrations across 12 physiochemical and experimental features.
- Trained 9 statistical models (regression, decision trees, ensemble methods) to rank feature importance and identify key interactions affecting NP tumor uptake.
- Presented research at the Houston Methodist Summer Science Symposium 2023 to 500+ attendees; received the Waggoner Foundation Merit Scholarship.
- Participated in a 7-person panel with TAMU EnMed students, advising high school students on medical and engineering career pathways.
- Received 30+ hours of bariatric surgery shadowing experience.

Research Intern – Environmental Exposures & Metabolic Disease

May 2024 – May 2025

Irma Rangel College of Pharmacy, Texas A&M University

College Station, TX

- Performed statistical analysis of publicly available NIH transcriptomic datasets to investigate associations between phthalates, microplastics, Type 2 Diabetes Mellitus, and obesity.
- Designed and executed full analysis workflow: data cleaning, normalization, differential gene expression analysis, pathway enrichment, and visualization.
- Wrote a 20-page research paper and delivered a 12-minute presentation at the 2024 Texas Junior Academy of Sciences, earning **3rd place** among hundreds of projects (Top 0.6% selectivity).

Research Intern – PCPG Cancer Transcriptomics Platform

May 2024 – Aug 2024

The University of Texas Health Science Center at San Antonio

San Antonio, TX

- Developed a unified, web-based analysis and visualization platform (**PCPG Analyzer**) for Pheochromocytoma and Paraganglioma (PCPG) transcriptomic datasets to support cancer researchers globally.
- Integrated interactive data exploration, differential expression analysis, and publication-quality visualizations using Python, Streamlit, and Pandas.
- Collaborated with graduate students from multiple institutions on data pipelines, statistical methods, and cloud deployment (Streamlit Cloud).
- Designed the platform to scale to additional cancer types, serving underfunded research teams worldwide.

Student Trainee – High Performance Scientific Computing

Jan 2024

Clarkson University (NSF-Funded Program)

Potsdam, NY (Remote)

- Participated in an NSF-funded, graduate-level training program as a high school junior, collaborating with graduate students from Stony Brook University and TU Dortmund.
- Solved eigenvalue problems and ordinary differential equations (ODEs) using PETSc (Portable, Extensible Toolkit for Scientific Computation) and SLEPc (Scalable Library for Eigenvalue Problem Computations).
- Implemented Proper Orthogonal Decomposition (POD) simulations to reduce computational intensity of scientific modeling tasks.

WORK EXPERIENCE

AI Systems / Site Reliability Engineer (SRE) Intern

May 2026 – Present

Lockheed Martin

Houston, TX (Remote)

- Architecting an AI agent system in partnership with the Splunk Delivery Team to automate event triage, alert correlation, and root cause analysis across production systems
- System generates actionable remediation recommendations and automated reports, reducing Mean Time to Recovery and manual incident investigation time for engineering teams.

Information Systems Student Assistant

October 2025 – Present

The Association of Former Students

College Station, TX

- Architecting automated, end-to-end Salesforce Flow pipelines to integrate Stripe payment data into Salesforce, converting raw transaction payloads into usable business entities that support Accounting reconciliation, financial reporting, and downstream analytics.

Site Reliability Engineer (SRE) Intern

May 2025 – Aug 2025

Lockheed Martin

Houston, TX (Remote)

- Automate Splunk app updates and installations using Ansible, targeting one of the largest Splunk deployments in the industry.

- Reduced total time spent on Splunk app operations by approximately 95% through automated workflows and repeatable playbooks.
- Decreased labor hours related to configuration management and disaster recovery by roughly 30% using Ansible automation.
- Integrated automation with AWS GovCloud, GitLab CI/CD, and Podman to standardize and scale deployments across multiple environments.

Mathematics Tutor

Aug 2021 – May 2025

Kumon North America, Inc.

College Station, TX

- Tutored students from elementary through high school, emphasizing conceptual understanding, critical thinking, and independent problem-solving in mathematics.
- Adapted instruction to diverse learning styles, helping students progress beyond grade level in mathematics and STEM subjects.

Mathematics & Science Tutor

Jun 2021 – May 2025

Self-Employed

College Station, TX

- Designed personalized lesson plans for 10+ students in mathematics, computer science, and STEM topics.
- Helped students master content above grade level; guided students through building their own personal computers to develop practical technical skills.

Textbook Contributor & Programmer

Jan 2020 – May 2025

MY MATH APPS

College Station, TX

- Co-developed online interactive calculus materials used by 100,000+ students annually at Texas A&M University, University of South Carolina, and Villanova University.
- Partnered with Dr. Phil Yasskin (Professor, TAMU Mathematics Department) on interactive Calculus I–III tutorials, exercises, and assessments.
- Formulated and implemented 500+ online exercises and interactive problems covering calculus topics (derivatives, integrals, sequences, series, multivariable calculus).
- Produced 100+ plots and animations using JavaScript (Node.js, jQuery), Maple, and SageMath for browser-based visualization of core calculus concepts.

TEACHING, OUTREACH & SERVICE

Mathematics Tutor

Aug 2025 – Present

WHIT Program

San Angelo, TX (Remote)

- Tutor foster care youth in mathematics by identifying specific knowledge deficits and implementing targeted remediation strategies to help students achieve grade-level competency.

Intensive Care Unit Volunteer

Aug 2025 – Present

Baylor Scott & White Health

College Station, TX

- Support ICU physicians and nursing staff during active patient care, including procedural assistance (retrieving and supplying equipment bedside) and indirect patient repositioning using draw sheets, and unit restocking.
- Accumulated 80+ hours of direct clinical exposure to critical care workflows, patient monitoring, and interdisciplinary care team dynamics.

Junior Counselor – SEE-MATH Honors Camp

Jul 2020 – Jul 2024

Department of Mathematics, Texas A&M University

College Station, TX

- Mentored gifted middle school students (grades 6–8) to promote advanced STEM education, with a focus on underrepresented and minority communities.
- Taught campers to develop digital animations and mathematical visualizations using the Maple Computer Algebra System (CAS).
- Website: <https://see-math.math.tamu.edu/>

Boys State Delegate

Mar 2024 – Jun 2024

American Legion Auxiliary – Texas Boys State

Austin, TX

- Selected as one of the top 1000 high school students in Texas (Top 0.23%) for the prestigious Texas Boys State program, a week-long intensive simulation of state government operations.
- Gained hands-on experience in legislative, executive, and judicial processes; developed critical thinking, public speaking, and teamwork skills through active participation in mock government.

- Engaged in leadership workshops focused on civic engagement, ethical decision-making, and collaborative governance.

Instructor – Computer Science Outreach

Jun 2022 – Jul 2022

Safe Tech Inc. (STI) & AggieSTEM Programs

College Station, TX

- Designed and programmed an I²C driver for the Raspberry Pi Sense HAT (HTS221 humidity sensor) for use with an ESP32 microcontroller for educational applications.
- Trained STEM teachers from Brownsville ISD in Python programming and Jupyter Notebooks as part of Project Lead-The-Way (PLTW), a “train-the-trainer” initiative for minority-serving institutions.
- Built and programmed autonomous rover vehicles using computer vision (CV) for K–12 STEM outreach.

Junior Counselor – Summer Computing Academy

Jun 2020 – Jul 2022

High Performance Research Computing, Texas A&M University

College Station, TX

- Mentored students in machine learning, Python programming, and cybersecurity topics at TAMU’s Summer Computing Academy, focusing on underrepresented and minority students.
- Featured in a local TV news segment (KBTX) to promote the camp and inspire younger students to pursue computing careers. Interview: bit.ly/CampSmartInterview2022 (Timestamp: 1:05)

Volunteer

Dec 2022

Brazos Valley Food Bank

College Station, TX

- Assisted with sorting items and assembling food boxes for distribution to local food pantries.

Instructor – Intro to Programming (Scratch)

Jun 2022

Aggieland Country School

College Station, TX

- Created and implemented a curriculum introducing elementary students (grades 1–4) to programming concepts using Scratch and Scratch Jr.

PROJECTS

PCPG Analyzer | *Python, Streamlit, Pandas, NumPy, Matplotlib*

Aug 2024

- Built a web-based framework for dynamic analysis and visualization of Pheochromocytoma & Paraganglioma (PCPG) transcriptome data to support cancer researchers worldwide.
- Deployed interactive visualizations and exploratory data analysis tools via Streamlit. Repository: github.com/aptitudepi/pcpg-analyzer

TIDALHack / Doctor-Robot | *Groq LLMs, XGBoost, Streamlit*

Oct 2024

- Developed a full-stack application that maps natural-language symptom descriptions to structured medical data and classifies into 40+ diseases using an XGBoost classifier.
- Integrated Groq-hosted LLMs (Llama models) with a machine learning classifier and Streamlit front-end. Repository: github.com/aptitudepi/TIDALHack

Mole Day Calculator | *Python, Streamlit*

Oct 2024

- Created a Streamlit web app for converting between moles and mass across all periodic table elements and compounds.
- Deployed as a public tool for chemistry education in celebration of Mole Day (10/23). Repository: github.com/aptitudepi/mole-day

US Cyber Games CTF Challenge | *LAMP Stack, SQL Injection, PHP*

Sep 2024

- Implemented a Capture-the-Flag style cybersecurity challenge exploiting an SQL injection vulnerability in a LAMP-based hacker forum.
- Demonstrated secure coding and remediation strategies alongside attack techniques for educational purposes.

TAMUHack – AI Voice MFA | *Raspberry Pi, ESP8266, Flask, CNN*

Jan 2024

- Led a 4-person team to build an AI-enabled multi-factor voice authentication system using Raspberry Pi SBC, ESP8266 microcontroller, and a CNN-based voice recognition model.
- Won “Best Beginner Hardware Hack” at TAMUHack 2024 (4th overall out of 1000+ participants). Repository: github.com/aptitudepi/2024TAMUHack

Hospital Plunge (TAMU Datathon) | *CNN, Flask, Keras, scikit-learn, Pandas*

Oct 2023

- Placed **3rd** out of 600+ participants for predicting patient hospital length of stay using convolutional neural networks.

- Implemented data cleaning, K-fold cross validation, PCA for dimensionality reduction, and regularization via dropout. Repository: github.com/aptitudepi/Hospital-Plunge

LinuxMAN | *Linux, Scripting*

Mar 2024

- Compiled a 2,537-page PDF reference of Linux manual pages to support training for Linux Operating System Fundamentals competitions.
- Repository: github.com/aptitudepi/LinuxMAN

Java Chess | *Java, Swing*

Mar 2024

- Implemented a fully functional chess game with a Java Swing GUI, enforcing legal moves, check/checkmate, and game rules.
- Repository: github.com/aptitudepi/UnCheckChess

ESP32 HTS221 I²C Driver | *C/C++, ESP32*

Sep 2022

- Developed an I²C driver interfacing the Raspberry Pi Sense HAT HTS221 humidity sensor with an ESP32 microcontroller for educational use.
- Repository: github.com/aptitudepi/esp32_hts221_i2c_driver

AWARDS & HONORS

- **Elkin Scholarship in CSCE**, Texas A&M University Department of Computer Science & Engineering.
- **James W & Lee Roach Endowed Scholarship**, Texas A&M University College of Engineering.
- **Southerland Aggie Leader Scholarship**, Texas A&M University – awarded for student involvement at Texas A&M University.
- **Dean's Honor Roll — Fall 2025, Spring 2026**, Texas A&M University College of Engineering – awarded for maintaining a GPA of at least 3.75 across 15+ credit hours.
- **Best Aggie Hack**, HowdyHack — 1st place hardware/software hack at Texas A&M, October 2025.
- **Brown Foundation Scholar (Craig C. and Galen T. Brown Foundation Scholarship)**, Texas A&M University – elite engineering scholarship recognizing the top 0.5% of incoming students.
- **President's Endowed Scholar (Mora Waddell Boone and James L. Boone, Sr. '21 Scholarship)**, Texas A&M University.
- **Dr. Amy Lum and Dr. Joseph Han '99 Dean's Scholars Award**, Texas A&M University College of Engineering.
- **Shubhra Kar LiFT Scholarship**, Linux Foundation.
- **Taylor Austin Broussard Cybersecurity Foundation Scholarship**, TAB Cyber Foundation. 2025 – 2027.
- **Hispanic Forum Scholarship**, The Hispanic Forum of Bryan-College Station.
- **Loyd Reddig '51 Endowed Memorial Scholarship**, Texas A&M University.
- **Nina Heard Astin Scholarship**, Nina Heard Astin Charitable Trust.
- **Gloria S. Sale/Bryan Educational Trust Scholarship**, Bryan Coca-Cola Educational Trust.
- **CSISD Education Foundation Pillar Scholarship** (sponsored by Amarillo National Bank), CSISD Education Foundation.
- **Oakwood Intermediate School PTO Scholarship**.
- **National Honor Society Semifinalist Scholarship**, National Association of Secondary School Principals.
- **National Merit Finalist**, National Merit Scholarship Corporation – 1 of 15,000 finalists selected from 1.3 million high school seniors nationwide.
- **National Merit Scholar**, National Merit Scholarship Corporation – awarded based on PSAT performance and academic record.
- **3rd Place**, Texas Junior Academy of Sciences (TJAS) 2024, Medicine & Health I – phthalates/microplastics and metabolic disease research (Top 0.6%).
- **National Cyber Scholar with Honors**, National Cyber Scholarship Foundation (2023, Top 0.0195%; 2024, Top 1.25%) – ranked among top 500 peers on GIAC GFACT certification exam both years.
- **3rd Place**, Linux Operating System Fundamentals, BPA National Leadership Conference 2024 (Top 0.0075% of 40,000).
- **5th Place**, Information Technology Concepts, BPA National Leadership Conference 2024 (Top 0.0125% of 40,000).
- **Best Beginner Hardware Hack**, TAMUHack 2024 – AI voice-based MFA system using Raspberry Pi and CNN (4th overall out of 1,000+ participants).

- **3rd Place**, TAMU Datathon 2023 Hospital Plunge challenge – CNN for patient hospital length-of-stay prediction (Top 0.5% of 600+).
- **Waggoner Foundation Merit Scholarship**, Houston Methodist Research Institute, 2023 – awarded at the Summer Science Symposium.
- **CyberPatriot National Semifinalist**, Air Force Association – Top 100 nationally in 2022 (Top 3.33%) and Top 150 in 2023.
- **Boys State Delegate**, American Legion Texas Boys State 2024 – selected among top 0.23% of Texas high school students.
- **AP Scholar with Distinction**, College Board – average ≥ 3.5 across all AP exams and scores ≥ 3 on 5+ exams.
- **Honorable Mention** (4th place), 2023 George H.W. Bush Presidential Library & Museum Essay Contest (Top 4% of 100+ applicants).
- **1st Prize Overall**, TAMU HPRC Summer Computing Academy 2022 – ML-based disease diagnosis from symptom input.
- **5th Prize Overall — as Mentor**, TAMU HPRC Summer Computing Academy 2021 – mentored high school team (as a middle school student) on cryptocurrency tweet sentiment analysis.
- **3rd Prize Overall**, TAMU HPRC Summer Computing Academy 2021 – cybersecurity model for hospital networks.
- **All-Region (1st Violin) — 3rd Chair**, TMEA Region 8 Orchestra, 2021.
- **1st Place — Hall of Fame**, NSA/NSF GenCyber Program 2019 – identified security flaws in the program platform.

PUBLICATIONS & PRESENTATIONS

- **Benchmarking Performance of Machine Learning Algorithms for Fluid Flow Simulation by Leveraging Cluster Computing: A Case Study for High-Performance Computing** – NASA Thermal and Fluids Analysis Workshop (Online), September 7, 2022. Presented Raspberry Pi cluster benchmarks using SLURM for IoT ML workloads. Slides and recording: go.nasa.gov/3VABkhv; youtu.be/famiRrm6Auk
- **MY Math Apps Calculus Online Materials** – Contributed interactive exercises, plots, and animations to MY MATH APPS Calculus I–II–III online resources used at Texas A&M University, University of South Carolina, and Villanova University (100,000+ students annually). Website: bit.ly/myma-calc-2023-db
- **Short Fiction in TEENS Publish Anthologies** – Stories “The Mind” (*The Isolated Generation*, 2020), “Anomaly” (*A Spark in the Void*, 2021), “Artificial” (*The Aftermath*, 2022), “Terminus” (*Stories Written in the Stars*, 2023), and “Breakdown” (*Thoughts on Paper*, 2024), published via Amazon KDP.

CERTIFICATIONS

- **GIAC Security Essentials Certification (GSEC)** – GIAC, May 2024.
- **GIAC Foundational Cybersecurity Technologies (GFACT)** – GIAC, August 2023.
- **TestOut Linux Pro Certification** – TestOut, April 2024.
- **Microsoft Certified: Azure Fundamentals** – Microsoft, May 2024.
- **Microsoft Certified: Azure AI Fundamentals** – Microsoft, May 2024.
- **Microsoft Certified: Security, Compliance, and Identity Fundamentals** – Microsoft, May 2024.
- **Cisco Certified Support Technician (CCST) – Cybersecurity & Networking** – Cisco, May 2024.
- **PMI Project Management Ready** – Project Management Institute, May 2024.
- **Master Certification in Entrepreneurship** – Certiport, May 2024.
- Full certification list (150+ credentials): bit.ly/BanerjeeDevkumarFullCertList

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, HTML/CSS, R, Bash, SQL, YAML
ML/AI Frameworks: PyTorch, PyTorch Lightning, TensorFlow, Keras, scikit-learn, XGBoost, Pandas, NumPy
Data & Scientific Tools: Pandas, NumPy, Matplotlib, SLURM, SageMath, Maple, Jupyter, PETSc, SLEPc
Cloud & DevOps: AWS GovCloud, Docker, Podman, Ansible, Git, GitLab CI/CD, GitHub Actions
Web Frameworks: Flask, Streamlit, Node.js, jQuery, RESTful APIs
Databases: SQL, PostgreSQL, MySQL, MongoDB
Systems & Other: Linux, ESP32 microcontroller, Raspberry Pi, 3D Printing, CAD, FASTA (data format)

LANGUAGES

Bengali: Native proficiency

English: Native proficiency

Spanish: Working proficiency

Hindi: Limited working proficiency